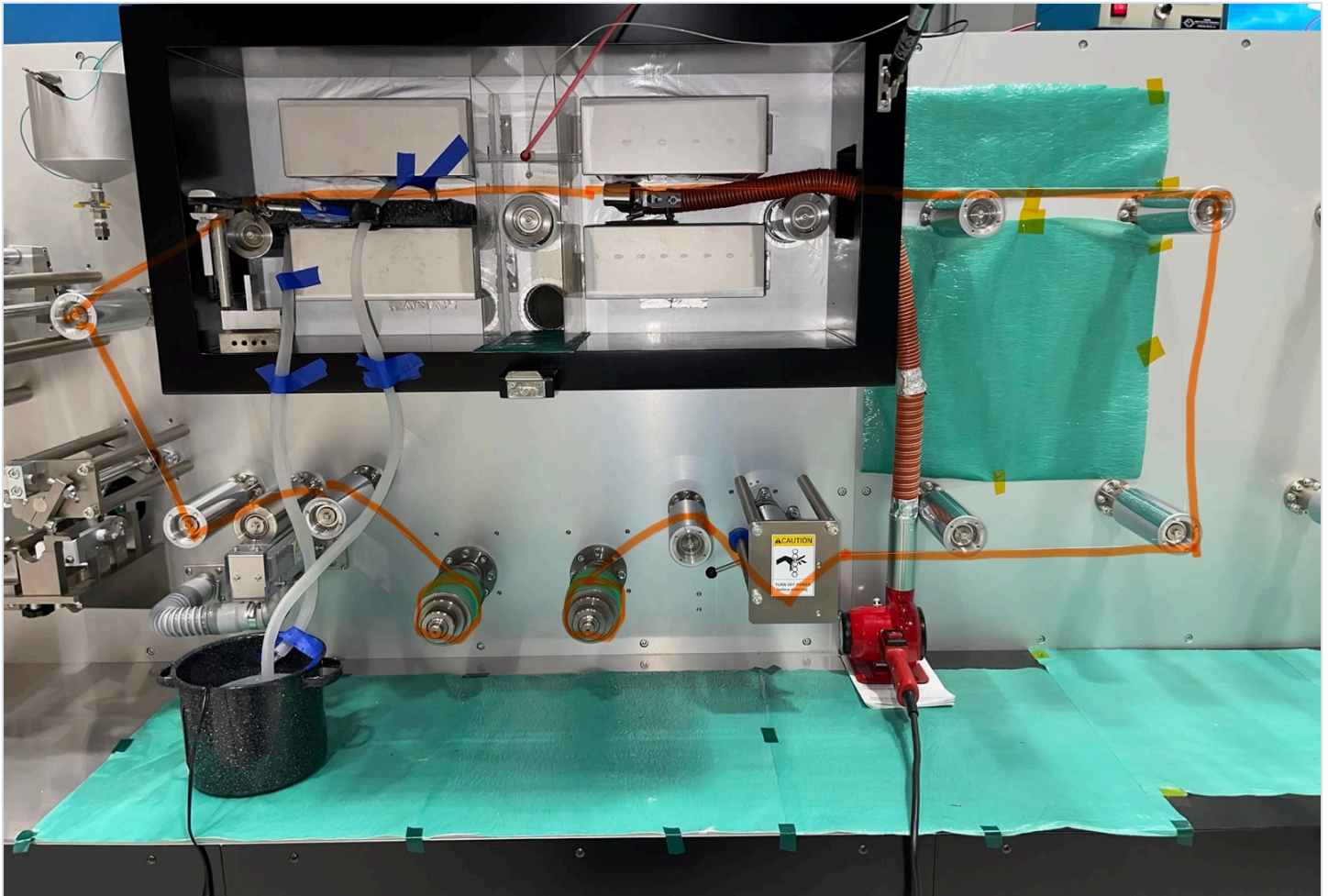


Kenneth Farhan

Rutgers University · B.S. Mechanical Engineering, cum laude · May 2026



Featured project: continuous roll-to-roll electrospay fiber coating: full system in operation, fiber path traced from unwind through deposition, heated drying, and rewind.

Winner: 2026 Senior Design Award, Best Mechanical Poster & Presentation

Rutgers School of Engineering, Department of Mechanical & Aerospace Engineering

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Functionalizing Fibers Using Electrospray Deposition

The problem

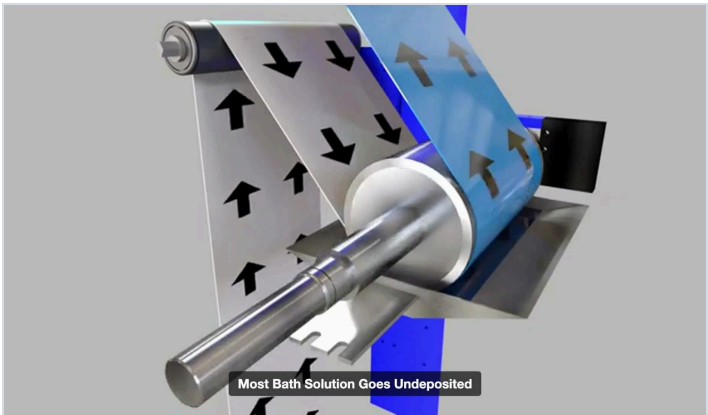
Industrial roll-to-roll (R2R) fiber coating uses dip baths: the fiber is pulled through an open solution reservoir, and most of the bath never ends up on the fiber. With specialty chemistries this is expensive, and contamination forces whole-bath disposal.

Our solution

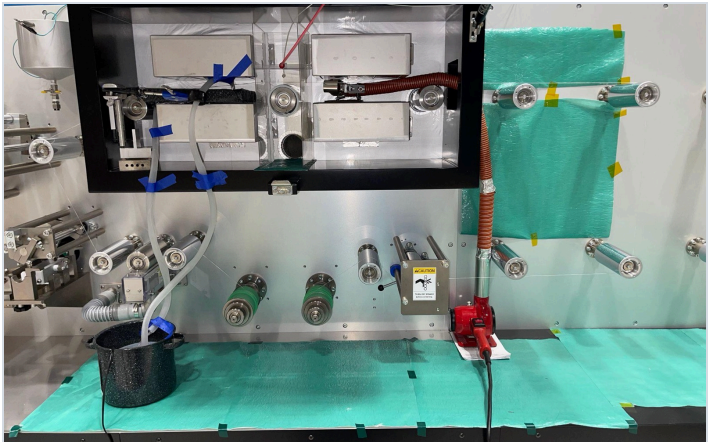
A modular electrospray deposition system that replaces the dip bath inside an existing Yasui Seiki R2R machine, with zero permanent modifications. A 15 kV potential atomizes polymer solution into charged nano-droplets that are electrostatically steered onto the grounded, moving fiber, depositing material only where it is wanted.

Validated result

A continuous one-hour run produced **6 m of uniformly coated fiber** under a stable Taylor cone-jet with zero high-voltage faults, the lab's first continuous electrospray manufacturing process.



From the team's process animation: in conventional dip coating, most of the bath solution goes undeposited.



The complete system during a run: syringe pump and high-voltage supply feeding the deposition enclosure, heated drying duct downstream.

6 m / 1 hr continuous coated fiber, validated run	15 kV stable Taylor cone-jet, zero faults	0 permanent modifications to host machine	4x CFD-verified airflow margin at needle	98% of \$700 budget utilized, on plan
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My role (4-person team)

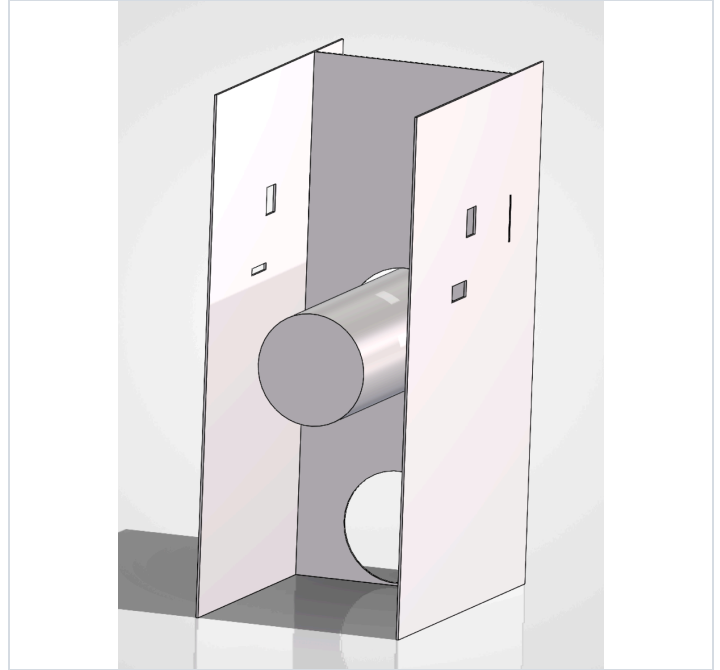
- **Simulation owner:** sole owner of the airflow modeling: ANSYS Fluent 3D conjugate heat transfer CFD, per ASME V&V 20.
- **CAD:** modeled the deposition enclosure, collector shroud, and fixtures in SolidWorks (23 parts, 2 assemblies, 6 engineering drawings).
- **High-voltage subsystem:** designed the 15 kV delivery and electrical-isolation enclosure; fabricated it via laser cutting and 3D printing.
- **Firmware:** rebuilt the syringe pump's Arduino control code from scratch (open source, github.com/kencarsick/syringe-pump-firmware).
- **Financial officer:** owned the budget and 13-item multi-vendor BOM.

Design & CAD

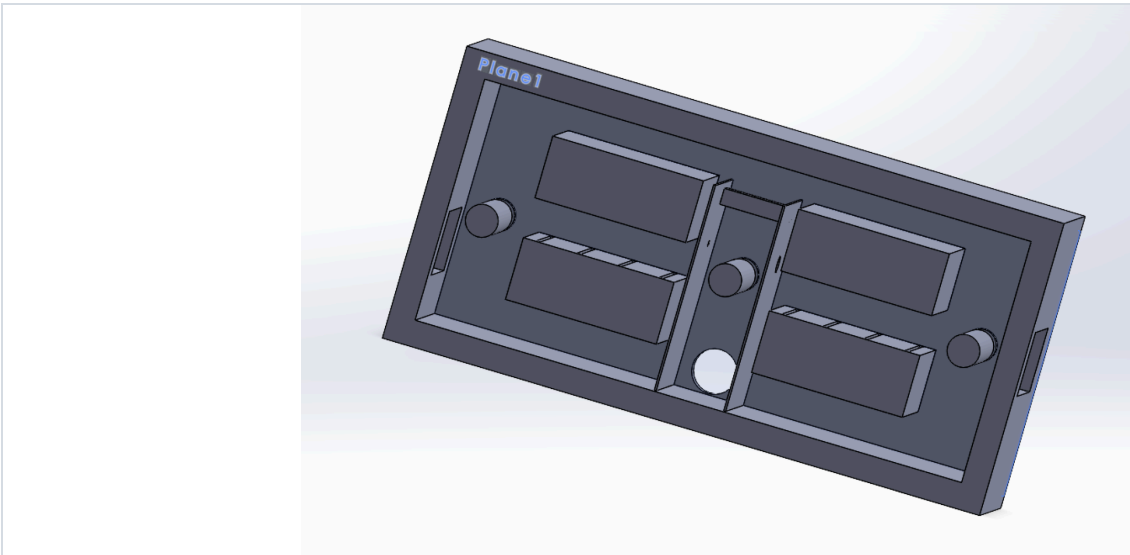
The deposition enclosure had to satisfy three hard constraints simultaneously: **electrical isolation** (all-insulating materials, ≥ 30 mm needle-to-metal clearance, no exposed conductors at 15 kV), a **tight spatial envelope** between the machine's four 3 kW air heaters and three steel rollers, and **airflow control** so heater exhaust could not disturb the spray. Design work ran from cardboard mockups dimensioned in situ, through a weighted decision-matrix material selection (cast acrylic over polycarbonate), to a fully detailed SolidWorks package released for laser cutting and 3D printing.



Low-cost first: a hand-annotated cardboard mockup fitted inside the R2R machine to capture real cutout dimensions before committing to CAD.



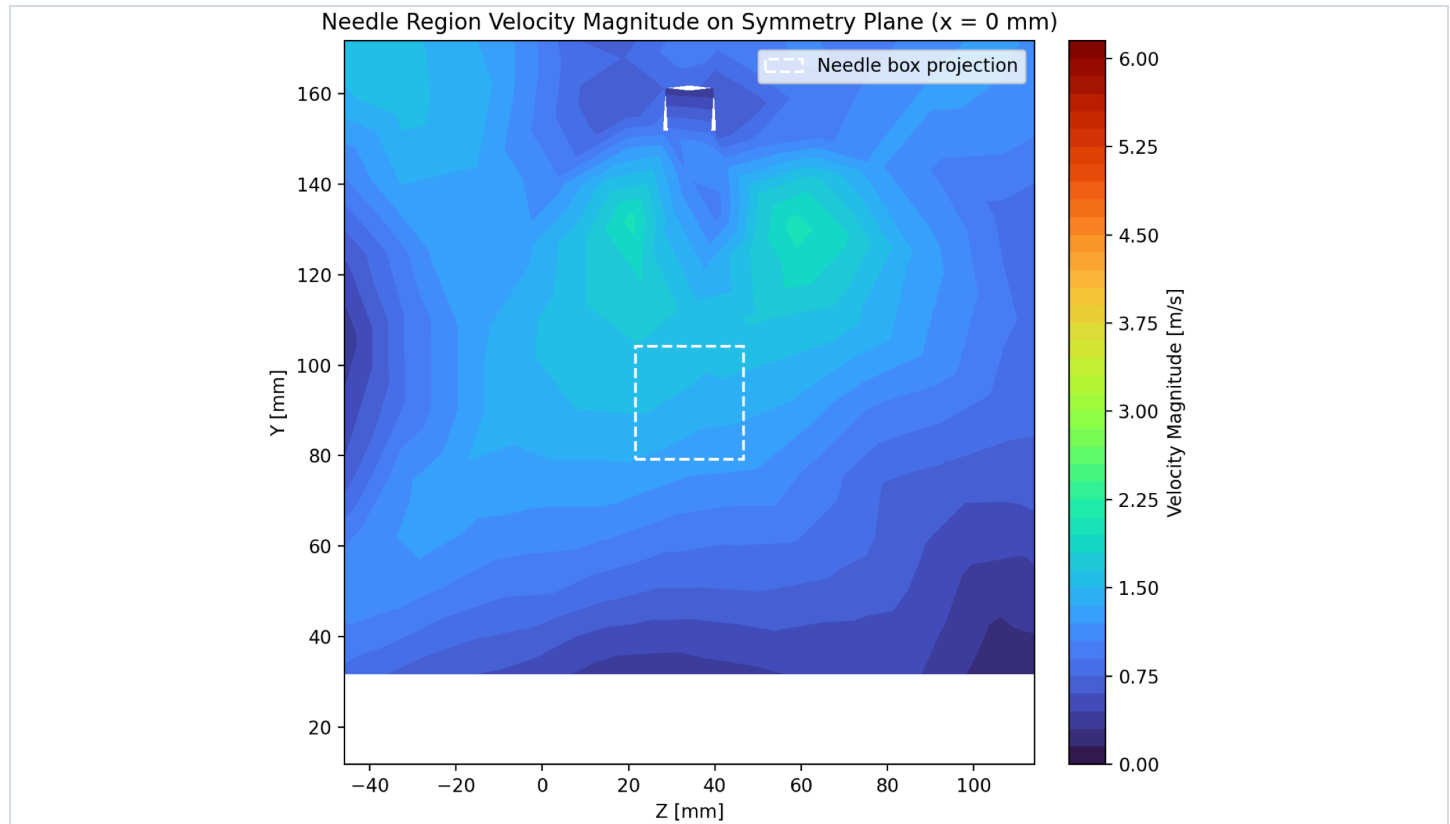
SolidWorks: collector shroud with the roller passing through slotted shield panels; fiber registers through laser-cut slots, no pulleys.



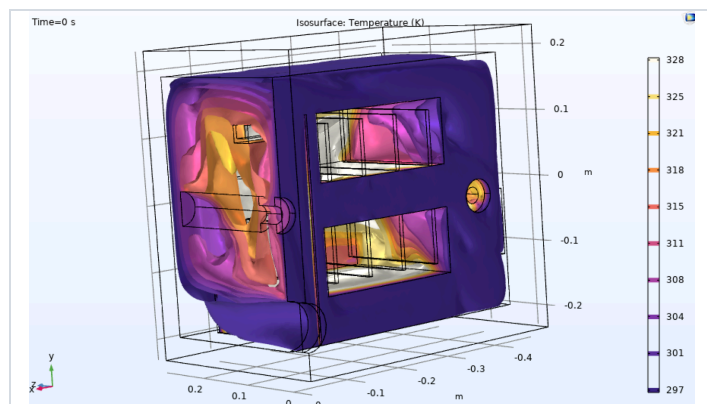
SolidWorks assembly of the deposition enclosure interior: insulating panel layout, spray divider, and roller pass-throughs. 23 parts, 2 assemblies, 6 drawings across the design package.

Simulation: CFD & Multiphysics

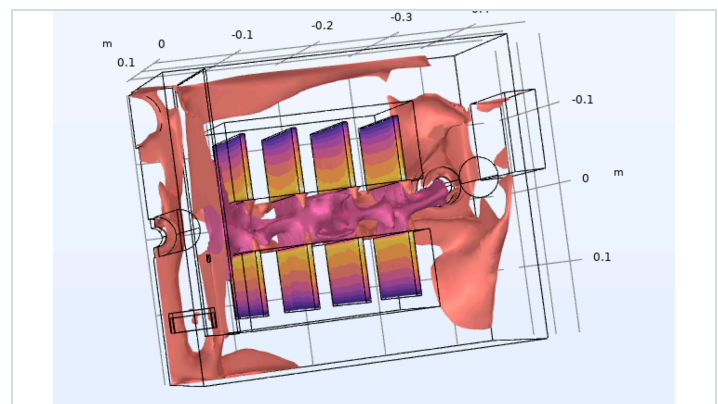
Electrospray fails if air moves faster than ~ 6 m/s at the needle: the Taylor cone-jet destabilizes. The enclosure sits directly in the path of four 3 kW air heaters, so I built a **3D conjugate heat transfer CFD model in ANSYS Fluent**: $k-\omega$ SST turbulence, energy equation with buoyant ideal-gas air, half-symmetry domain, $\sim 336,000$ polyhedral cells, heaters as measured velocity/temperature inlets) to verify the design before fabrication, following ASME V&V 20 verification and validation practice.



Result: velocity-magnitude contours on the symmetry plane with the needle-region box overlaid, showing 1.24 m/s average / 1.58 m/s peak at the needle in the conservative all-heaters-on case: a $\sim 4\times$ margin below the spray-disruption threshold. The as-built system ran a stable cone-jet for a full hour.



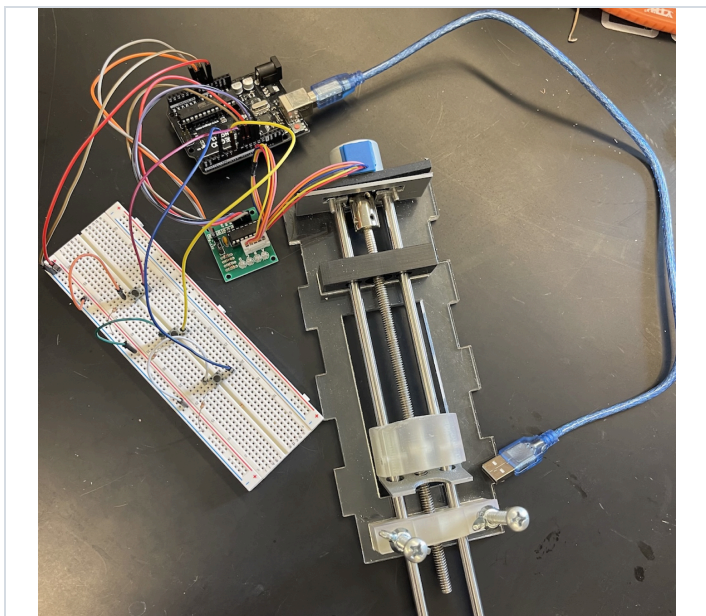
COMSOL Multiphysics thermal study (fall design phase): temperature isosurfaces inside the enclosure used to size the drying zone.



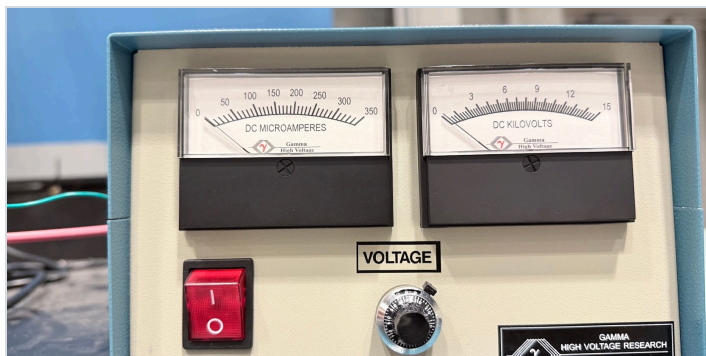
COMSOL airflow isosurfaces from the same study, superseded in spring by the higher-fidelity Fluent conjugate heat transfer model.

Build, Electronics & Firmware

Fabrication used laser-cut cast acrylic (chosen partly because laser-cutting polycarbonate off-gasses chlorine), drilled-and-tapped polycarbonate fixtures, 3D-printed PLA parts with print orientation chosen for load direction, and adhesive bonding with slotted-edge registration; no conductive fasteners anywhere near the field.



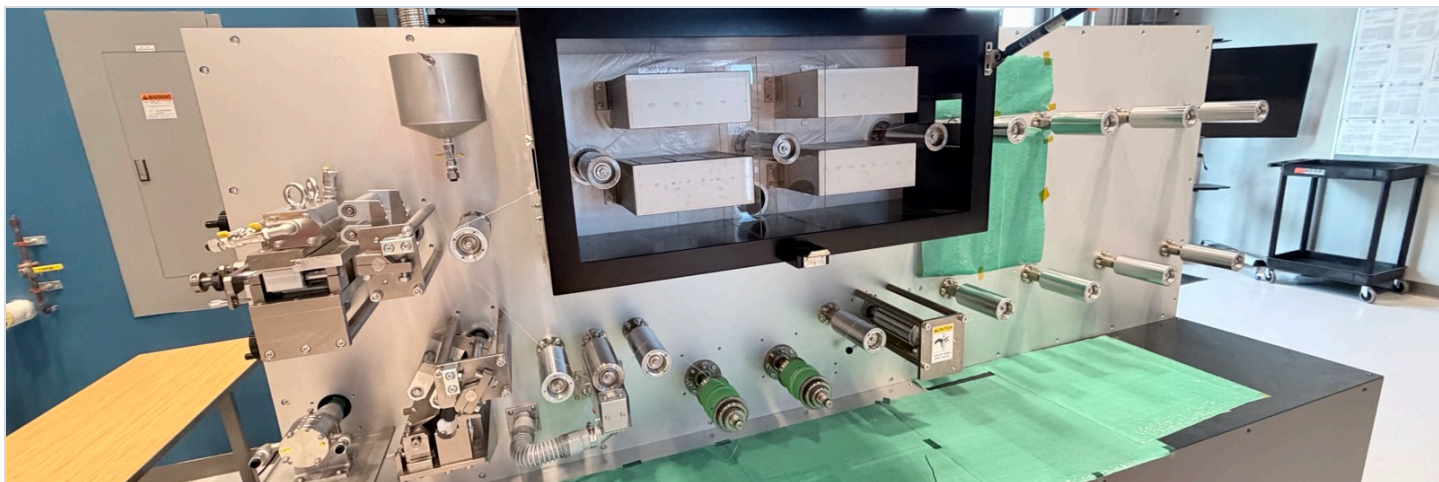
Custom syringe pump: Arduino + stepper on a lead screw with 3D-printed carriage. I rebuilt its inherited, undocumented firmware into 620 lines of non-blocking C++ (five-state machine, serial command protocol, EEPROM persistence) so it accepts syringe diameter and flow rate like a commercial lab pump. **Open source:** github.com/kencarsick/syringe-pump-firmware



Gamma High Voltage 0-15 kV DC supply. I designed the delivery path (bolt-and-clip electrode, ≥ 30 mm clearances, UHMW-sleeved roller to kill parasitic ground attraction) with NFPA 70E-informed procedures.



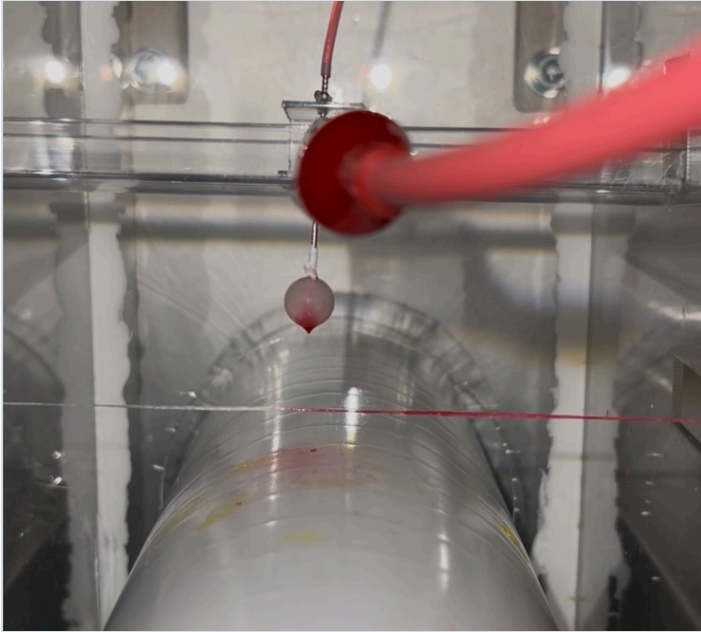
The isolation enclosure installed: insulating panels, acrylic spray shields, and the sleeved collector roller; a fully reversible retrofit.



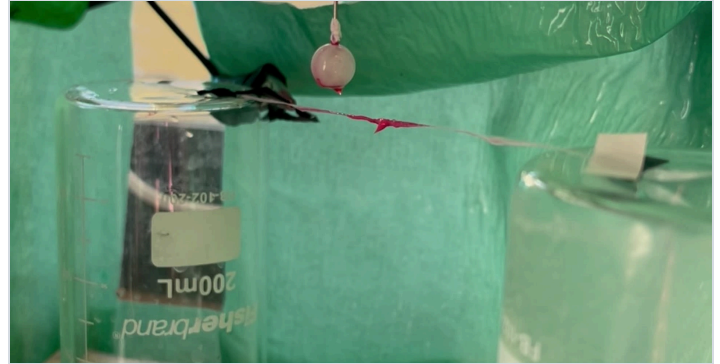
As-built: the deposition enclosure integrated into the Yasui Seiki roll-to-roll coater in the Rutgers Advanced Manufacturing Lab.

Operation & Validation

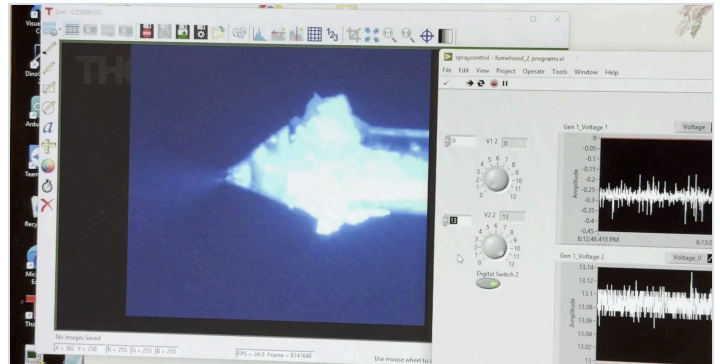
Dry nylon fiber is a dielectric: the spray has no ground path. Our saturation zone wets the fiber in a closed regenerative water loop immediately upstream, dropping its effective surface resistivity by ~ 9 orders of magnitude so the moving fiber inherits ground. The validated end-to-end run: 0.1 m/min line speed, 2.5 mL/hr flow, 15 kV, one hour, 6 m of uniformly coated fiber, coating intact after rewind.



Inside the enclosure mid-run: charged needle with pendant droplet above the sleeved collector roller, fiber line crossing the deposition zone.



Coating transfer made visible: dyed solution depositing onto the moving fiber strand for full-length coverage verification.



Process monitoring: live microscope feed of the spray plume during characterization, alongside the spray-control interface.

Failure modes found and engineered out

- Heater airflow destabilizing the cone-jet $\rightarrow$ diagnosed empirically, then bounded with the Fluent CFD model and enclosure redesign.
- Parasitic charge attraction to the grounded steel roller $\rightarrow$ UHMW polyethylene sleeve and revised clearances.

Recognition & Other Experience



2026 Senior Design Award: Best Mechanical Poster & Presentation, presented to Kenneth Farhan. Rutgers School of Engineering, Dept. of Mechanical & Aerospace Engineering.

HVAC / MEP design, Premier Engineering Group (Summer 2025)

Mechanical/HVAC engineering intern at an MEP consultancy: designed and drafted chilled-water, VRF, and hydronic heating/cooling systems in AutoCAD across commercial projects; ran Wrightsoft load calculations and zoning analyses for equipment sizing; conducted field surveys and as-built documentation for boiler/chiller plants, clean-room ventilation, and BMS integrations.

Academic foundation

GPA 3.31, cum laude, Dean's List $\times$ 5. Straight A's senior year in Heat Transfer, Power Plants, Aerospace Materials, and Design & Manufacturing I-II.

Toolbox

SolidWorks	AutoCAD	ANSYS Fluent CFD	COMSOL
MATLAB / Simulink	C++ / Arduino	Python	3D printing
Laser cutting	GD&T	Wrightsoft	Git

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Full design reports, CAD package, firmware source, and process video available on request.